TITLE

Design and apply context-aware access controls based on user identity, device posture, location, and application risk.

OVERVIEW OF THE PROJECT

- In today's digital world, traditional security models are no longer effective due to cloud computing, remote work, and mobile device usage. Organizations face increasing cyber threats, making it essential to implement advanced security mechanisms.

- This project focuses on implementing a **context-aware access control system** based on Zero Trust architecture. Instead of trusting users based on network location, the system evaluates multiple parameters such as user identity, device security posture, geographic location, and application risk before granting access.

- The solution uses dynamic decision-making to either allow, restrict, or deny access. For example, a trusted employee accessing from a secure device will get full access, while access from an unknown location may require multi-factor authentication or may be blocked.

- This approach improves security, reduces attack surface, and ensures safe access to organizational resources.

LIST OF VULNERABILITIES

S.NO	VULNERABILITY NAME	CWE NO
1	Cross-Site Scripting (Stored)	79
2	Broken Access Control	285
3	SQL Injection	89
4	Broken Authentication	285
5	IDOR	639
6	Security Misconfiguration	732
7	CSRF	352
8	Cleartext Transmission	319
9	Clickjacking	1021

SAMPLE REPORT (ONE EXAMPLE)

VULNERABILITY NAME:

Cross-Site Scripting (Stored)

CWE:

79

OWASP:

Injection

DESCRIPTION:

This vulnerability occurs when malicious scripts are injected into web applications and executed in users' browsers.

BUSINESS IMPACT:

Attackers can steal user data, perform actions on behalf of users, and compromise sensitive information.

STAGE 2

OVERVIEW (NESSUS UNDERSTANDING)

- Nessus is a vulnerability scanning tool used to identify security issues in systems and networks. It helps organizations detect vulnerabilities such as misconfigurations, weak passwords, and outdated software.
- The scanning process includes defining targets, configuring scan policies, running scans, identifying vulnerabilities, and generating reports. Nessus categorizes vulnerabilities based on severity levels like low, medium, and high.
- This tool helps security teams prioritize risks and take corrective actions to improve system security.

TARGET DETAILS

- Target Website: testfire.net
- Target IP: 54.83.41.36

VULNERABILITY TABLE

S.NO	VULNERABILITY	SEVERITY	PLUGIN
1	XSS	Low	Web Scanner
2	Broken Access Control	Medium	SSH weak HMAC
3	SQL Injection	Medium	N/A
4	Broken Authentication	Low	HSTS Missing
5	IDOR	Info	Service Detection

SAMPLE REPORT

VULNERABILITY:

SQL Injection

SEVERITY:

Medium

PORT:

80/443

DESCRIPTION:

SQL Injection allows attackers to manipulate database queries using malicious input.

SOLUTION:

- Use parameterized queries
- Validate inputs

BUSINESS IMPACT:

Data breaches, financial loss, and reputation damage.

STAGE 3

TITLE

Security Operations Center (SOC) and SIEM Implementation

SOC

SOC is a centralized unit that monitors and manages security threats in an organization. It detects, analyzes, and responds to cyber incidents.

SOC Cycle

1. Monitor
2. Detect
3. Analyze
4. Respond
5. Recover

SIEM

SIEM (Security Information and Event Management) collects and analyzes log data from different systems to detect threats.

SIEM Cycle

- Data Collection
- Correlation
- Analysis
- Alerting
- Response

MISP

MISP is a threat intelligence platform used to share cyber threat data between organizations.

THREAT INTELLIGENCE

It involves collecting and analyzing information about cyber threats to prevent attacks.

INCIDENT RESPONSE

It is the process of identifying, managing, and mitigating cyber attacks.

QRadar

QRadar is a SIEM tool used for monitoring security events and detecting threats in real time.

CONCLUSION

Stage 1:

Learned about web application vulnerabilities and access control.

Stage 2:

Understood Nessus scanning and vulnerability assessment.

Stage 3:

Learned SOC, SIEM, and security monitoring.

FUTURE SCOPE

- Advanced threat detection
- Automation using AI
- Improved security monitoring